



MAARULA CLASSES

TARGET- NIMCET / CUET.PG By: Amit Katiyar (MCA-JNU)

**APPLICATION OF
DERIVATIVES**

NIMCET PYQ

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1. $f(x) = \sin x - \tan x, x \in \left(0, \frac{\pi}{2}\right)$, then tangent at any point on the curve:

NIMCET 2007

- (a) below the curve (b) above the curve
(c) on the curve (d) None of these

2. If $f(x) = \max(2x + 1, 3 - 4x)$. Then the minimum value of $f(x)$

NIMCET 2007

- (a) $1/3$ (b) $5/3$ (c) $3/2$ (d) None

3. A is 8 miles east of B, C is 10 miles north of B, D is 13 miles east of C, E is 2 mile north of D. Then the shortest distance between A and E is: (AOD)

- (a) 13 (b) 12 (c) 5 (d) none

NIMCET 2007

4. A point on the parabola $y^2 = 18x$ at which the ordinate increases at twice the rate of the abscissa is: (AOD)

- (a) $\left(-\frac{9}{8}, \frac{9}{2}\right)$ (b) $(2, -4)$ (c) $(2, 4)$ (d) $\left(\frac{9}{8}, \frac{9}{2}\right)$

NIMCET 2007

5. The function $f(x) = 2 \sin x + \sin 2x, x \in [0, 2\pi]$ has absolute maximum and minimum at

NIMCET 2008

- (a) $\frac{\pi}{3}, \frac{5\pi}{3}$ (b) $\frac{\pi}{3}, \pi$ (c) $\frac{5\pi}{3}, \pi$ (d) None

6. Water runs into a conical tank of radius 5 ft and height 10 ft at a constant rate of 2 ft/min. how fast is the water level rising when the water is 6 ft deep?

NIMCET 2009

- (a) $\frac{2}{9}$ ft/min (b) $\frac{2}{9\pi}$ ft/min
(c) $\frac{2\pi}{9}$ ft/min (d) $\frac{\pi}{9}$ ft/min

7. An open top box is to be made out of a piece of cardboard measuring $6m \times 6m$ by cutting off equal squares from the corners and turing up the sides. The height of the box maximum volume is

NIMCET 2009

- (a) 2.0 m (b) 2.5 m (c) 1.2 m (d) None

8. The rate of increase of length of the shadow of a man 2 m height, due to a lamp at 10 m height, when he is moving away from it at the rate of 2 m/sec is

NIMCET 2010

- (a) $1/2$ m/sec (b) $2/5$ m/sec
(c) $1/3$ m/sec (d) 5 m/sec

9. If $y = a \log x + bx^2 + x$ has its extremum value at $x = -1$ and $x = 2$, then

NIMCET 2010

- (a) $a = 2, b = -1$ (b) $a = -2, b = \frac{1}{2}$
(c) $a = 2, b = -\frac{1}{2}$ (d) $a = 1, b = -\frac{1}{2}$

10. The minimum value of $px + qy$, where $xy = r^2$ is

- (a) $2r\sqrt{pq}$ (b) $2pq\sqrt{r}$ (c) $-2r\sqrt{pq}$ (d) $\sqrt{pqr}$

11. Normal to the curve $y = x^3 - 3x + 2$ at the point $(2, 4)$ is

NIMCET 2012

- (a) $9x - y - 14 = 0$ (b) $x - 9y + 40 = 0$
(c) $x + 9y - 38 = 0$ (d) $-9x + y + 22 = 0$

12. The point on the curve $y = 6x - x^2$, where the tangent is parallel to X-axis is

NIMCET 2012

- (a) $(0, 0)$ (b) $(2, 8)$ (c) $(6, 0)$ (d) $(3, 9)$

13. The function x^x decreases in the interval

NIMCET 2012

- (a) $(0, e)$ (b) $(0, 1)$ (c) $\left(0, \frac{1}{e}\right)$ (d) None

14. Find the point at which the tangent to the curves $y = \sqrt{4x - 3} - 1$ has its slope $\frac{2}{3}$.

NIMCET 2013

- (a) $(3, 3)$ (b) $(3, 2)$ (c) $(2, 3)$ (d) $(2, 2)$

15. The minimum value of the function $y = 2x^3 - 21x^2 + 36x - 20$ is

NIMCET 2013

- (a) -120 (b) -126 (c) -128 (d) None

16. A condition that $x^3 + ax^2 + bx + c$ may have no extremum is

NIMCET 2014

- (a) $a^2 \geq 3b$ (b) $b^2 < 3a$
(c) $a^2 < 3b$ (d) $b^2 \geq 3a$

17. Two non-negative numbers whose sum is 9 and the product of the one number and square of the other number is maximum, are

NIMCET 2014

- (a) 5 and 4 (b) 3 and 6
(c) 1 and 8 (d) 7 and 2

18. The equation of the tangent at any point of curve $x = a \cos 2t, y = 2\sqrt{2}a \sin t$ with m as its slope is

NIMCET 2015

- (a) $y = mx + a\left(m - \frac{1}{m}\right)$ (b) $y = mx - a\left(m + \frac{1}{m}\right)$
(c) $y = mx + a\left(a + \frac{1}{a}\right)$ (d) $y = amx + a\left(m - \frac{1}{m}\right)$

19. A function $F : (0, \pi) \rightarrow R$ defined by $f(x) = 2 \sin x + \cos 2x$ has

NIMCET 2015

- (a) a local minimum but no local maximum
(b) a local maximum but no local minimum
(c) both local minimum and local maximum
(d) neither a local minimum nor a local maximum





20. Let $f(x) = x^2 - bx + c$, b is an odd positive integer. If $f(x) = 0$, has two prime numbers 1 as roots and $b + c = 35$, then the global minimum value of $f(x)$ is

NIMCET 2016

- (a) $-\frac{183}{4}$ (b) $\frac{173}{16}$ (c) $-\frac{81}{4}$ (d) $\frac{17}{2}$

21. Area of the greatest rectangle that can be inscribed in ellipse is

NIMCET 2016

- (a) $\sqrt{ab}$ (b) $2ab$ (c) ab (d) $\frac{a}{b}$

22. The critical point and nature find the function $f(x, y) = x^2 - 2x + 2y^2 + 4y - 2$ is

NIMCET 2017

- (a) (1, 1) maximum (b) (1, -1) maximum
(c) (1, 1) minimum (d) (1, -1) minimum

23. If the radius of a circle change at the rate of $\frac{-2}{\pi}$ m/s at what rate does the circle's area change when the radius is 10 m?

NIMCET 2018

- (a) $40 \text{ m}^2/\text{s}$ (b) $30 \text{ m}^2/\text{s}$
(c) $-30 \text{ m}^2/\text{s}$ (d) $-40 \text{ m}^2/\text{s}$

24. The tangent at the point (2, -2) to the curve $x^2y^2 - 2x = 4(1 - y)$ does not pass through the point

NIMCET 2019

- (a) (-2, -7) (b) (-4, -9)
(c) $(4, \frac{1}{3})$ (d) (8, 5)

25. Find the interval(s) on which the graph $y = 2x^3e^x$ is increasing

NIMCET 2020

- (a) $(-3, 0)$ and $(0, \infty)$ (b) $(-\frac{3}{2}, 0)$ and $(0, \infty)$
(c) $(-3, \infty)$ only (d) None of these

26. The function $f(x) = \frac{x}{1+x \tan x}$, $0 \leq x \leq \frac{\pi}{2}$ is maximum when

NIMCET 2021

- (a) $x = \sec x$ (b) $x = \tan x$
(c) $x = \cos x$ (d) None of these

27. A particle is at rest at the origin. It moves along the X-axis with an acceleration $x - x^2$, where x is the distance of the particle at time t . the particle next comes to rest after it has covered a distance

NIMCET 2022

- (a) 1 (b) $1/2$ (c) $3/2$ (d) 2

28. If $\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$, ($a > b$) and $x^2 - y^2 = c^2$ cut at right angles, then

NIMCET 2022

- (a) $a^2 + b^2 = 2c^2$ (b) $b^2 - a^2 = 2c^2$
(c) $a^2 - b^2 = 2c^2$ (d) $a^2 - b^2 = c^2$

29. The equation of the tangent at any point of curve $x = a \cos 2t$, $y = 2\sqrt{2} \sin t$ with m as its slope is

NIMCET 2022

- (a) $y = mx + a\left(m - \frac{1}{m}\right)$ (b) $y = mx - a\left(m + \frac{1}{m}\right)$
(c) $y = mx + a\left(a + \frac{1}{a}\right)$ (d) $y = amx + a\left(m - \frac{1}{m}\right)$

30. Between any two real roots of the equation $e^x \sin x = 1$, then equation $e^x \cos x = -1$ has

NIMCET 2023

- (a) Atleast one root (b) Exactly one root
(c) No root (d) None of these

31. The maximum value of $f(x) = (x - 1)^2(x + 1)^3$ is equal to $\frac{2^p 2^q}{3125}$ then the ordered pair of (p, q) will be

NIMCET 2023

- (a) (3, 7) (b) (7, 3)
(c) (5, 5) (d) (4, 4)

32. Consider the function $f(x) =$

$$\begin{cases} -x^3 + 3x^2 + 1, & \text{if } x \leq 2 \\ \cos(x) & \text{if } 2 < x \leq 4 \\ e^{-x} & \text{if } x > 4 \end{cases}$$

Which of the following statement about $f(x)$ is true (maxima minima)

- (a) $f(x)$ has a local maximum at $x = 1$, which is also the global maximum
(b) $f(x)$ has a local maximum at $x = 2$, which is not the global maximum.
(c) $f(x)$ has a local maximum at $x = \pi$, but it is not the global maximum.
(d) $f(x)$ has a global maximum at $x = 0$.

NIMCET 2024

33. Consider the function $f(x) = x^{2/3}(6-x)^{1/3}$. Which of the following statement is FALSE? (AOD)

- (a) f is increasing in the interval (0, 4)
(b) f is decreasing in the interval (6, ∞)
(c) f has a point of inflection at $x = 0$
(d) f has a point of inflection at $x = 6$

NIMCET 2024

34. If the line $a^2x + ay + 1 = 0$, for some real number a , is normal to the curve $xy = 1$, then (AOD)

- (a) $a < 0$ (b) $0 < a < 1$
(c) $a > 0$ (d) $-1 < a < 1$

NIMCET 2024



35. The slope of the normal line to the curve $x = t^2 + 3t - 8$ and $y = 2t^2 - 2t - 5$ at the point $(2, -1)$ is

NIMCET 2025

- (a) $\frac{22}{7}$ (b) -5
(c) $-\frac{6}{7}$ (d) $-\frac{7}{6}$

36. The curve $y = \frac{x}{1 + x \tan x}$ attains maxima

NIMCET 2025

- (a) at $x = \cos x$
(b) at $x = -\cos x$
(c) at $x = \cos x$ and for some $x \in [-2\pi, \pi]$
(d) for some $x \in [-2\pi, \pi]$

ANSWER KEY

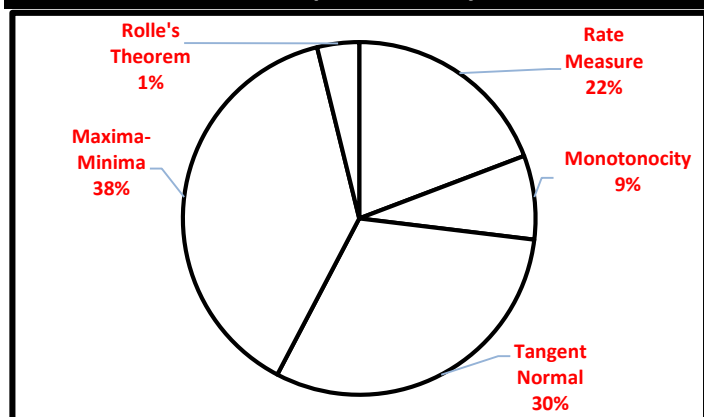
1.	b	2.	b	3.	a	4.	d	5.	a
6.	b	7.	d	8.	a	9.	c	10.	a
11.	c	12.	d	13.	c	14.	b	15.	c
16.	c	17.	b	18.	b	19.	c	20.	c
21.	b	22.	d	23.	d	24.	a	25.	a
26.	c	27.	c	28.	c	29.	b	30.	a
31.	b	32.	B	33.	c	34.	a	35.	d
36.	a								

Analysis 2007 to 2025

Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	04	NIMCET 2016	01
NIMCET 2008	01	NIMCET 2017	01
NIMCET 2009	02	NIMCET 2018	01
NIMCET 2010	02	NIMCET 2019	01
NIMCET 2011	01	NIMCET 2020	01
NIMCET 2012	03	NIMCET 2021	01
NIMCET 2013	02	NIMCET 2022	02
NIMCET 2014	03	NIMCET 2023	03
NIMCET 2015	02	NIMCET 2024	03
		NIMCET 2025	02

Sub Topicwise Analysis

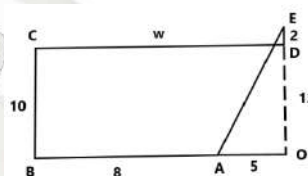


Sub Topicwise Questions Asked in NIMCET

Sub Topic	Question Asked
Rate Measure	05
Monotonocity	02
Tangent Normal	08
Maxima-Minima	10
Rolle's Theorem	01

SOLUTION

1. (b)
2. (b)
3. (b)



Shortest distance between A and E

$$= \sqrt{5^2 + (12)^2}$$

$$= \sqrt{169}$$

$$AE = 13$$

4. (d)

Given Parabola: $y^2 - 18x$

Differentiate w.r.t. parameter t (or directly w.r.t. x):

$$2y \frac{dy}{dx} - 18$$

$$\frac{dy}{dx} - \frac{9}{y}$$

So slope

$$\frac{dy}{dx} - \frac{9}{y}$$

"ordinate increases at twice the rate of abscissa" means

$$\frac{dy}{dx} = 2$$

Solve for y

$$\frac{9}{y} = 2$$

$$y = \frac{9}{2}$$

From $y^2 - 18x$:

$$\left(\frac{9}{2}\right)^2 - 18x$$

$$\frac{81}{4} - 18x$$

$$x = \frac{81}{72} = \frac{9}{8}$$

$$\text{So point} = \left(\frac{9}{8}, \frac{9}{2}\right)$$

5. (a) Given, $f(x) = 2\sin x + \sin 2x$

$$\Rightarrow f'(x) = 2\cos x + 2\cos 2x$$

For max or min of $f(x)$

$$\Rightarrow f'(x) = 0 \Rightarrow 2\cos^2 x + \cos x - 1 = 0$$

$$\Rightarrow \cos x = \frac{-1 \pm \sqrt{1+8}}{4} = -1, \frac{1}{2}$$

Critical points are $\pi, \frac{\pi}{3}, \frac{5\pi}{3}$

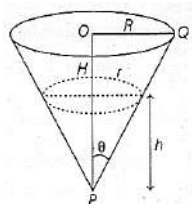


$$f(\pi) = 0, f\left(\frac{\pi}{3}\right) = \frac{3\sqrt{3}}{2}, f\left(\frac{5\pi}{3}\right) = -\frac{3\sqrt{3}}{2}$$

Function has absolute maxima at $x = \frac{\pi}{3}$

Absolute minima at $x = \frac{5\pi}{3}$

6. (b) $R = 5, H = 10$



$$\tan \theta = \frac{R}{H} = \frac{r}{h}$$

$$\frac{5}{10} = \frac{r}{h} \Rightarrow \frac{1}{2} = \frac{r}{h} \Rightarrow h = 2r$$

$$\frac{dh}{dt} = 2 \frac{dr}{dt}$$

Given,

$$h = 6 \Rightarrow 6 = 2r \Rightarrow r = 3$$

$$V = \frac{1}{3} \pi r^2 h$$

$$\frac{dV}{dt} = \frac{\pi}{3} \left[2hr \frac{dr}{dt} + r^2 \frac{dh}{dt} \right]$$

$$2 = \frac{\pi}{3} \left[2 \times 6 \times 3 \frac{dr}{dt} + 9 \frac{dh}{dt} \right]$$

$$\frac{2}{\pi} = \left[6 \frac{dh}{dt} + 3 \frac{dh}{dt} \right] \text{ as } \frac{dh}{dt} = 2 \frac{dr}{dt}$$

$$\frac{2}{\pi} = 9 \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{2}{9\pi} \text{ ft/min}$$

7. (d) Let the side of square which is cut from the corners = x

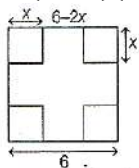
Then volume of box, $V = (6 - 2x)^2 \cdot x$

$$\Rightarrow \frac{dV}{dx} = \frac{d}{dx} (36 + 4x^2 - 24x)x$$

$$\Rightarrow \frac{dV}{dx} = 36 + 12x^2 - 48x = 0$$

$$\Rightarrow x^2 - 4x + 3 = 0$$

$$\Rightarrow (x - 3)(x - 1) = 0$$



Critical points are $x = 3, 1$

$$\frac{d^2V}{dx^2} = 0 + 24x - 48$$

$$\frac{d^2V}{dx^2} \Big|_{x=1} = 24 - 48 = -24 < 0 \text{ (maxima)}$$

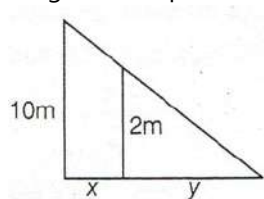
$$\text{And, } \frac{d^2V}{dx^2} \Big|_{x=3} = 72 - 48 = 24 > 0 \text{ (minima)}$$

Volume is maximum when height is 1 m.

8. (a) Height of man = 2 m

Moving away from lamp = 2 m/sec

Height of lamp = 10 m



Let x be the length between man and lamp and y be the length of shadow.

$$\frac{dx}{dt} = 2 \text{ m/sec (given)}$$

From figure:

$$\frac{10}{x+y} = \frac{2}{y} \rightarrow \text{Congruent triangles}$$

$$\Rightarrow 10y = 2x + 2y$$

$$\Rightarrow 8y = 2x \Rightarrow x = 4y$$

$$\Rightarrow \frac{dx}{dt} = \frac{4dy}{dt}$$

$$\Rightarrow \frac{1}{4} \times 2 = \frac{dy}{dt} \Rightarrow \frac{dy}{dt} = \frac{1}{2} \text{ m/sec}$$

9. (c) $y = a \log x + bx^2 + x$ has extremum at $x = -1$ and $x = 2$

$$f(x) = a \log x + bx^2 + x$$

$$f'(x) = \frac{a}{x} + b \times 2x + 1$$

Now, at critical point $x = -1$

$$f'(-1) = \frac{a}{-1} + 2b(-1) + 1 = 0$$

$$-a - 2b + 1 = 0$$

$$1 - 2b = a$$

Again, at $x = 2$

$$f'(2) = \frac{a}{2} + 2b \times 2 + 1 = 0$$

$$\frac{a}{2} + 4b + 1 = 0$$

From Eqs. (i) and (ii)

$$a = 2, b = -\frac{1}{2}$$

10. (a) The minimum value of $px + qy$, when $xy = r^2$

$$xy = r^2 \Rightarrow y = \frac{r^2}{x}$$

$$f(x) = px + qy = px + \frac{qr^2}{x}$$

$$f'(x) = p + \frac{qr^2}{x^2}(-1)$$

$$f'(x) = p + \frac{qr^2}{x^2}(-1) = 0$$

$$\Rightarrow x = \pm r \sqrt{\frac{q}{p}}$$

$$f''(x) = \frac{2qr^2}{x^3}$$

$$f''\left(r \sqrt{\frac{q}{p}}\right) > 0$$

Hence, minima at $x = \left(r \sqrt{\frac{q}{p}}\right)$

$$f(x) = px + \frac{qr^2}{x} \text{ at } x = r \sqrt{\frac{q}{p}} \Rightarrow f(x) = 2r\sqrt{pq}$$

11. (c) Normal to the curve $y = x^3 - 3x + 2$ at point (2,4)

$$y = x^3 - 3x + 2$$

$$\frac{dy}{dx} = 3x^2 - 3$$

$$\frac{dy}{dx} \Big|_{(2,4)} = 3 \times 2^2 - 3 = 9$$

Slope of tangent = 9 = m_1

Now, slope of normal (m_2) = $-1/\text{Slope of tangent } (m_1)$

$$m_2 = \frac{-1}{9}$$



Equation of normal at (2,4)

$$y - 4 = \frac{-1}{9}(x - 2)$$

$$\Rightarrow 9y - 36 = -x + 2$$

$$\Rightarrow x + 9y = 38$$

$$\Rightarrow x + 9y - 38 = 0$$

- 12. (d)** The point on the curve $y = 6x - x^2$, where tangent is parallel to X -axis.

Any line parallel to X -axis, then slope of the line $m = 0$

$$y = 6x - x^2$$

$$\frac{dy}{dx} = 6 - 2x = 0$$

$$6 = 2x \Rightarrow x = 3$$

(because of parallel to X -axis)

Now, put $x = 3$ at the curve,

$$y = 6 \times 3 - 9 = 9 \text{ (point on the curve} = (3,9))$$

- 13. (c)** Let $f(x) = x^x = y$

Any function decreases, then its derivative $f'(x) < 0$

$$y = x^x$$

$$\log_e y = x \log_e x \quad [x > 0]$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{x}{x} + \log_e x$$

$$\frac{dy}{dx} = x^x(1 + \log_e x) < 0$$

$$1 + \log_e x < 0 \quad [x^x > 0]$$

$$\log_e x < -1$$

$$\text{or } x < \frac{1}{e} \quad (x > 0)$$

$$x \in \left(0, \frac{1}{e}\right)$$

- 14. (b)** Point on the curve $y = \sqrt{4x-3} - 1$

has tangent is slope $= \frac{2}{3}$

$$\frac{dy}{dx} = \frac{2}{3}$$

$$y = \sqrt{4x-3} - 1$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{4x-3}} \times 4 = \frac{2}{3}$$

$$\sqrt{4x-3} = 3 \Rightarrow 4x-3 = 9 \Rightarrow x = 3$$

Put $x = 3$ on the curve, we get $y = 2$

$$(x, y) = (3, 2)$$

- 15. (c)** The minimum value of the function

$$f(x) = 2x^3 - 21x^2 + 36x - 20 \text{ is}$$

$$f'(x) = 6x^2 - 42x + 36 = 0$$

$$(x-6)(x-1) = 0$$

$x = 6, 1$ are critical point

$$f''(x) = 12x - 42$$

$$f''(6) = 12 \times 6 - 42 = 30 > 0 \rightarrow \text{Minima}$$

$$f''(1) = 12 \times 1 - 42 = -30 < 0 \rightarrow \text{Maxima}$$

There is minima at $x = 6$

$$f_{\min} = f(6) = 2 \times 6^3 - 21 \times 36 + 36 \times 6 - 20 = -128$$

- 16. (c)** A condition that $x^3 + ax^2 + bx + c$ has no extremum.

A function has no maxima and no minima when it is increasing or decreasing everywhere.

$$f(x) = x^3 + ax^2 + bx + c$$

$$f'(x) = 3x^2 + 2ax + b$$

Now, let $f'(x) > 0$, for every $x \in R$

$$3x^2 + 2ax + b > 0$$

Any quadratic equation is always positive when

coefficient of x^2 is positive and discriminant is negative.

$$3 > 0 \text{ and } D = b^2 - 4ac < 0$$

$$(2a)^2 - 4 \times 3b < 0$$

$$4a^2 - (3b)4 < 0$$

$$a^2 - 3b < 0$$

$$a^2 < 3b$$

- 17. (b)** Let two numbers be x and y .

Then, $x \geq 0, y \geq 0, x + y = 9, x^2y \rightarrow \text{maximum}$

$$y = 9 - x$$

$$x^2(9 - x) = f(x)$$

$$f(x) = 9x^2 - x^3 \Rightarrow f'(x) = 18x - 3x^2 = 0$$

$$= 3x(6 - x) = 0$$

$x = 0, 6$ are two critical points

$$f''(x) = 18 - 6x$$

$$f''(0) = 18 - 6 \times 0 = 18 \rightarrow \text{minima}$$

$$f''(6) = 18 - 6 \times 6 = -18 \rightarrow \text{maxima}$$

$$\therefore x = 6$$

From Eq. (i), $y = 3$

- 18 (b)** Equation of tangent with slope m

$$x = \cos 2t$$

$$y = 2\sqrt{2}a \sin t$$

$$\frac{dx}{dt} = -2a \sin 2t \text{ and } \frac{dy}{dt} = 2\sqrt{2}a \cos t$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2\sqrt{2}a \cos t}{-2a \sin 2t}$$

$$= \frac{\sqrt{2} \cos t}{-2 \sin t \cos t} = \frac{-1}{\sqrt{2} \sin t} = m \text{ (given)}$$

$$\sin t = \frac{-1}{\sqrt{2}m}$$

Then,

$$x = \cos 2t$$

$$y = 2\sqrt{2}a \sin t$$

$$= a(1 - 2\sin^2 t)$$

$$y = 2\sqrt{2}a \times \left(\frac{-1}{\sqrt{2}m}\right)$$

$$= a \left(1 - 2 \times \left(\frac{-1}{\sqrt{2}m}\right)^2\right)$$

$$y = \frac{-2a}{m}$$

$$x = a \left(1 - \frac{1}{m^2}\right)$$

Then, equation of tangent-

$$y - y_1 = m(x - x_1)$$



$$y + \frac{2a}{m} = m \left(x - a \left(1 - \frac{1}{m^2} \right) \right)$$

$$y + \frac{2a}{m} = mx - am + \frac{a}{m}$$

$$y = mx - a \left(m + \frac{1}{m} \right)$$

19. (c) A function $F: (0, \pi) \rightarrow R$ defined

by $f(x) = 2\sin x + \cos 2x$ has

$$f'(x) = 2\cos x - 2\sin 2x = 0$$

$$\Rightarrow 2\cos x - (2\cos x \sin x) = 0$$

$$\Rightarrow 2\cos x(1 - \sin x) = 0 \Rightarrow \cos x = 0$$

$$\Rightarrow x = \frac{\pi}{2}$$

And $\sin x = \frac{1}{2}$

$$\Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$f''(x) = -2\sin x - 4\cos 2x$$

$$f''\left(\frac{\pi}{2}\right) = -2 \times 1 - 4\cos 2 \times \frac{\pi}{2}$$

$$= -2 - 4 \times (-1) = 2 \rightarrow \text{minima}$$

$$f''\left(\frac{\pi}{6}\right) = -2 \times \frac{1}{2} - 4\cos 2 \times \frac{\pi}{6}$$

$$= -1 - 4 \times \frac{1}{2} = -3 \rightarrow \text{maxima}$$

$f(x)$ has both local minima and local maxima.

20. (c) $f(x) = x^2 - bx + c$, b is an odd positive integer.

$f(x) = 0$ has prime number as roots and $b + c = 35$.

Let α, β be the roots of $x^2 - bx + c = 0$

Hence, $\alpha + \beta = b$, $\alpha \times \beta = c$

As $\alpha + \beta = b$, b is an odd integer

Sum of two numbers is odd when one is even and other is odd number.

As we know α, β are prime numbers, so α must be 2 for given conditions.

$$f(2) = 4 - 2b + c = 0$$

$$= 2b - c = 4$$

$$b + c = 35$$

From Eqs. (i) and (ii)

$$b = 13, c = 22$$

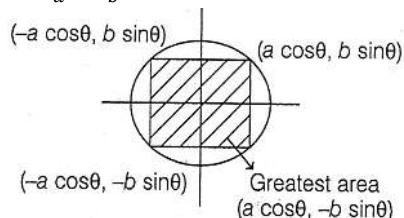
$$f(x) = x^2 - 13x + 22$$

$$f'(x) = 2x - 13 = 0 \Rightarrow x = \frac{13}{2}$$

There is only one critical point.

$$\text{It has } f_{\min} = f\left(\frac{13}{2}\right) = \left(\frac{13}{2}\right)^2 - 13 \times \left(\frac{13}{2}\right) + 22 = \frac{-81}{4}$$

21. (b) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



Area of rectangle = $l \times b$

$$= 2a \cos \theta \times 2b \sin \theta$$

$$= 2ab \sin 2\theta$$

For greatest $\sin 2\theta = 1$ ($-1 \leq \sin \theta \leq 1$) $\Rightarrow \theta = \frac{\pi}{4}$

Area of greatest rectangle = $2ab$

22. (d) Critical point and nature for function

$$f(x, y) = x^2 - 2x + 2y^2 + 4y - 2$$

This is extrema for two variables, hence we use partial differentiation,

$$\text{Partial derivatives } \frac{\partial f}{\partial x} = 2x - 2 = 0$$

$$\Rightarrow x = 1 \text{ and } \frac{\partial f}{\partial y} = 4y + 4 = 0 \Rightarrow y = -1$$

So, critical points are $(1, -1)$

$$f''(x) = 2 > 0 \rightarrow \text{minima}$$

$$f''(y) = 4 > 0 \rightarrow \text{minima}$$

So, $f(x, y)$ has minimum at $(1, -1)$

23. (d) Radius of circle change at $\frac{-2}{\pi}$ m/s

$$\frac{dA}{dt} = ? \text{ of circle } r = 10 \Rightarrow \frac{dr}{dt} = \frac{-2}{\pi} \text{ m/s}$$

Area of circle $(A) = \pi r^2$

$$\Rightarrow \frac{dA}{dt} = \pi \times 2r \frac{dr}{dt}$$

$$\Rightarrow \frac{dA}{dt} = \pi \times 2 \times 10 \times \frac{-2}{\pi} = -40 \text{ m}^2/\text{sec}$$

$$\Rightarrow \frac{dA}{dt} = -40 \text{ m}^2/\text{sec}$$

24. (a) Curve $x^2y^2 - 2x = 4(1 - y)$,

$$2x^2y \frac{dy}{dx} + 2xy^2 - 2 = -\frac{dy}{dx} \times 4$$

$$\Rightarrow 2x^2y \frac{dy}{dx} + \frac{4dy}{dx} = 2 - 2xy^2 \Rightarrow \frac{dy}{dx} = \frac{2-2xy^2}{2x^2y+4} = \frac{1-xy^2}{x^2y+2}$$

$$\left. \frac{dy}{dx} \right|_{(2,-2)} = \frac{1-2 \times (-2)^2}{(2^2)(-2)+2} = \frac{7}{6}$$

$$\text{Tangent } y + 2 = \frac{7}{6}(x - 2)$$

$$\Rightarrow 6y + 12 = 7x - 14 \Rightarrow 7x - 6y = 26$$

Now, $7x - 6y = 26$

Tangent at $(2, -2)$ passes through $(8, 5)$, $(-4, -9)$ and $(4, 1/3)$

and not passes through $(-2, -7)$.

25. (a) $y = 2x^3e^x$

$$\frac{dy}{dx} = 6x^2e^x + 2x^3e^x > 0 = 2x^2e^x(x + 3) > 0, x > -3$$

But strictly increasing $x \neq 0$

$$x \in (-3, 0) \cup (0, \infty)$$

26. (c) $f(x) = \frac{x}{1+x \tan x}, 0 \leq x \leq \frac{\pi}{2}$

$$f'(x) = \frac{1 \cdot (1+x \tan x) - x(0 + \tan x + x \sec^2 x)}{(1+x \tan x)^2}$$

$$f'(x) = 0$$

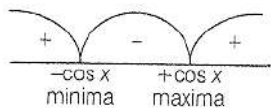


$$\Rightarrow \frac{1+x \tan x - x \tan x - x^2 \sec^2 x}{(1+x \tan x)^2} = 0$$

$$\Rightarrow 1 - x^2 \sec^2 x = 0 \Rightarrow \cos^2 x - x^2 = 0$$

$$\Rightarrow (\cos x - x)(\cos x + x) = 0$$

Critical point $x = \pm \cos x$



$f(x)$ has maxima at $x = \cos x$

27. (c) $a = x - x^2$, a is acceleration

$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2} = x - x^2$$

Here, x is the distance and v is velocity.

At $t = 0$, $u = 0$

Where, u is initial velocity,

$$\frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt}$$

$$\Rightarrow x - x^2 = v \frac{dv}{dx}$$

$$\Rightarrow v \frac{dv}{dx} = x - x^2$$

On integrating Eq. (i)

$$\int v dv = \int (x - x^2) dx \Rightarrow \frac{v^2}{2} = \frac{x^2}{2} - \frac{x^3}{3}$$

For $v = 0$,

$$3x^2 - 2x^3 = 0 \Rightarrow x = 0, 3/2$$

Hence, answer is $x = 3/2$ because particle start from rest.

28. (c) If two curves intersect orthogonally,

$$p_1 x^2 + q_1 y^2 = 1 \text{ and } p_2 x^2 + q_2 y^2 = 1$$

$$\text{Then, } \frac{1}{p_1} - \frac{1}{p_2} = \frac{1}{q_1} - \frac{1}{q_2}$$

$$\text{Hence, } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\text{Here, } p_1 = \frac{1}{a^2}, q_1 = \frac{1}{b^2}$$

$$\text{and for } x^2 - y^2 = c^2 \Rightarrow \frac{x^2}{c^2} - \frac{y^2}{c^2} = 1$$

$$\text{Hence, } p_2 = \frac{1}{c^2}, q_2 = -\frac{1}{c^2}$$

Putting these values in Eq. (i)

$$a^2 - c^2 = b^2 + c^2 \Rightarrow a^2 - b^2 = 2c^2$$

29. (b) Refer to solution 14.

30. (a) $e^x \sin x = 1$.

$$f(x) = \sin x - e^{-x}$$

Let it has 2 real roots α, β .

$$f'(x) = \cos x + e^{-x} = 0$$

Using Rolle's theorem-

There will be atleast one real root of its derivative between α and β

atleast one $c \in (\alpha, \beta)$, $f'(c) = 0$

$$e^{-c} + \cos c = 0$$

$$1 + e^c \cos c = 0 = g(c)$$

$$g(x) = e^x \cos x + 1$$

$$a < c < b$$

Atleast one real root of $g(x)$ between two real roots of $f(x)$.

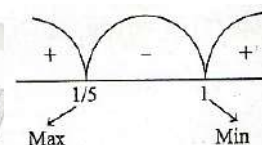
31. (b) $f(x) = (x-1)^2(x+1)^3$

$$f'(x) = 2(x-1)(x+1)^3 + 3(x+1)^2(x-1)^2$$

$$= (x-1)(x+1)^2[2(x+1) + 3(x-1)] = 0$$

$$\Rightarrow (x-1)(x+1)^2[5x-1] = 0$$

$$\Rightarrow x = 1, -1, 1/5$$



$$f\left(\frac{1}{5}\right) = \left(\frac{1}{5}-1\right)^2 \left(\frac{1}{5}+1\right)^3$$

$$= \left(-\frac{4}{5}\right)^2 \left(\frac{6}{5}\right)^3$$

$$= \frac{16}{5^2} \times \frac{6^3}{5^3} = \frac{2^7 \times 3^3}{5^5}$$

$$\text{Hence, } p = 7, q = 3$$

$$32. f(x) = \begin{cases} -x^3 + 3x^2 + 1 & x \leq 2 \\ \cos x & 2 < x < 4 \\ e^{-x} & x > 4 \end{cases}$$

$$f'(x) = \begin{cases} -3x^2 + 6x & x < 2 \\ -\sin x & 2 < x < 4 \\ -e^{-x} & x > 4 \end{cases}$$

$f'(x) = 0$ is local maximum

$$-3x^2 + 6x = 0$$

$$f''(x) = -6x + 6$$

$$f''(2) = -6 > 0$$

at $x = 2$ is local maximum

$$-\sin x = 0$$

$$x = \pi$$

$$= -\cos \pi = +1 \rightarrow \min$$

so $f(x)$ has local maximum at $x = 2$ which is not global max.

33. (c) $f(x) = x^{2/3} \cdot (6-x)^{1/3}$

$$f'(x) = x^{2/3} \cdot \frac{1}{3} \cdot (6-x)^{-2/3} + \frac{2}{3} (6-x)^{1/3} x^{-1/3}$$

$$f'(x) = x^{-1/3} \cdot (6-x)^{-2/3} \cdot \left(-\frac{x}{3} + \frac{2}{3}(6-x)\right)$$

$$f'(x) = \frac{1}{x^{1/3}(x-6)^{2/3}} \left(-\frac{x}{3} + 4 - \frac{2x}{3}\right)$$

$$f'(x) = -\frac{1}{x^{1/3}(x-6)^{2/3}} (x-4)$$

f has a point of inflection at $x = 0$

34. (c)

$$a^2x + ay + 1 = 0 \text{ for real no. of } a, \text{ is normal to the curve}$$

$$xy = 1$$



$$xy = 1$$

$$x \frac{dy}{dx} + y = 0$$

$$\frac{dy}{dx} = \frac{-y}{x}$$

Normal

$$\frac{-dy}{dx} = \frac{x}{y} = \frac{x}{1/x} = x^2$$

$$y = ax - \frac{1}{a}$$

$$x^2 = -a$$

$$a < 0$$

35. (d)

How to solve

Find the value of t at the given point, then calculate the slope of the tangent, and finally determine the slope of the normal. [3]

1. Find the value of t

Substitute the given point $(2, -1)$ into the equations for x and y .

$$2 = t^2 + 3t - 8$$

$$-1 = 2t^2 - 2t - 5$$

$$\text{Solve } t^2 + 3t - 10 = 0 \text{ to get } (t + 5)(t - 2) = 0, \text{ so } t = 2$$

$$\text{or } t = -5$$

$$\text{Solve } 2t^2 - 2t - 4 = 0 \text{ (or } t^2 - t - 2 = 0) \text{ to get } (t - 2)(t + 1) = 0,$$

$$\text{so } t = 2 \text{ or } t = -1$$

The common value of t is $t = 2$.

2. Calculate $\frac{dx}{dt}$ and $\frac{dy}{dt}$:

$$\text{Differentiate } x \text{ with respect to } t: \frac{dx}{dt} = 2t + 3.$$

$$\text{Differentiate } y \text{ with respect to } t: \frac{dy}{dt} = 4t - 2$$

3. Calculate the slope of the tangent line

$$\text{Use the formula } = \frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

$$\text{Substitute } t = 2 \text{ into the derivatives: } \frac{dy}{dx} = \frac{4(2) - 2}{2(2) + 3}$$

$$\frac{dy}{dx} = \frac{8 - 2}{4 + 3} = \frac{6}{7}.$$

4. Calculate the slope of the normal line

The slope of the normal line is the negative reciprocal of the tangent's slope.

$$\text{Slope of normal} = -\frac{1}{\text{slope of tangent}} = -\frac{1}{\frac{6}{7}}.$$

$$\text{Slope of normal} = -\frac{7}{6}.$$

36. (a)

$$y = \frac{x}{1 + x \tan x}$$

$$y' = \frac{1 - x^2 \sec^2 x}{(1 + x \tan x)^2} = 0$$

$$\Rightarrow x^2 = \cos^2 x$$

$$\Rightarrow x = \pm \cos x$$

$$\text{Let } N(x) = \cos^2 x - x^2. \text{ Then } N'(x) < 0$$

so N goes $+$ $\rightarrow$ $-$ and y' goes $+$ $\rightarrow$ $-$ $\Rightarrow$ maximum.

At $x = -\cos x$ it gives a minimum.

Hence maxima at $x = \cos x$